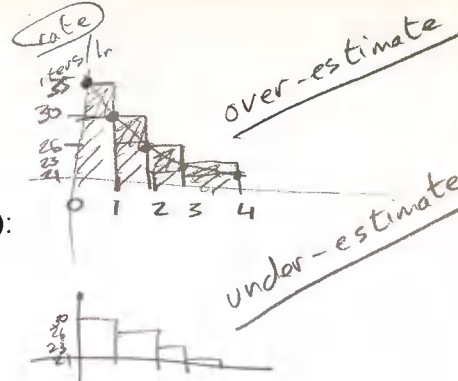


Section 3.1 How Do We Measure Distance Traveled (or Spilled Oil)

Assumption: Oil is leaking out of a container at a **decreasing rate**.

A. Leakage rate measured at **hourly** intervals (at the **beginning** of each hour):

Time (hours)	0	1	2	3	4
Rate (liters/hour)	35	30	26	23	21



1. Suppose we want to estimate the total amount of oil that has leaked out of the container during the four-hour time interval.

2. What is the maximum amount of oil that could have leaked out during the first **hour** (i.e., between $t=0$ and $t=1$)? the minimum? During subsequent hours?

During Hour Ending at $t=$	1	2	3	4
Maximum Leakage (liters)	35	30	26	23
Minimum Leakage (liters)	30	26	23	21

$$= 114$$

$$= 100$$

3. During the entire four-hour interval, what is the maximum amount of oil that could have leaked out?

4. What is the minimum amount?

5. What is the difference between these maximum and minimum values?

6. Where have we used the assumption that the rate of leaking is decreasing?

7. If you had to guess how much oil actually leaked during the four-hour interval, what would be your guess?

8. What is the maximum possible error in your guess? (i.e., what is the maximum possible difference between your guess and the true value?)

B. Leakage rate measured **every half-hour**.

Time (hours)	0	0.5	1	1.5	2	2.5	3	3.5	4
Rate (liters/hour)	35	33	30	27	26	24	23	22	21

1. What is the maximum amount of oil that could have leaked out during the first **half hour** (i.e., between $t=0$ and $t=0.5$)? the minimum? During subsequent half hour intervals?

During Half Hour Ending at $t=$	0.5	1	1.5	2	2.5	3	3.5	4
Maximum Leakage (liters)	17.5	16.5	15	13.5	13	12	11.5	11
Minimum Leakage (liters)	16.5	15	13.5	13	12	11.5	11	10.5

$$= 110.2$$

$$= 103.2$$

$$\frac{33}{2}$$

2. During the entire four-hour interval, what is the maximum amount of oil that could have leaked out? *110 l*
3. What is the minimum amount? *103 l*
4. What is the difference between these maximum and minimum values? *7 l* $\frac{1}{2}$ of previous
5. If you had to guess how much oil actually leaked during the four-hour interval, what would be your guess? *106.5 l*
6. What is the maximum possible error in your guess? (i.e., what is the maximum possible difference between your guess and the true value?) *3.5 l*

C. Shorter Time Intervals Between Measurements

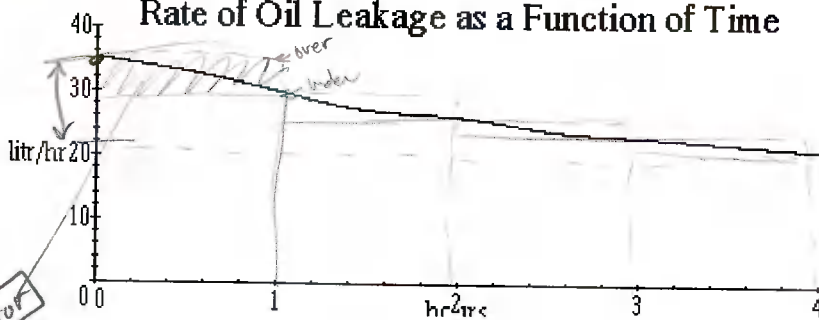
1. Suppose we could take measurements every $\frac{1}{10}$ hour? What would be the difference between the upper and lower estimates for the total leakage? *$14 \frac{l}{hr} \times \frac{1}{10} hr = 1.4 l$*
2. Suppose we could take measurements every $\frac{1}{100}$ hour? What would be the difference between the upper and lower estimates for the total leakage? *0.14 l*

3. What can you conclude about the accuracy to which you can estimate the total leakage?

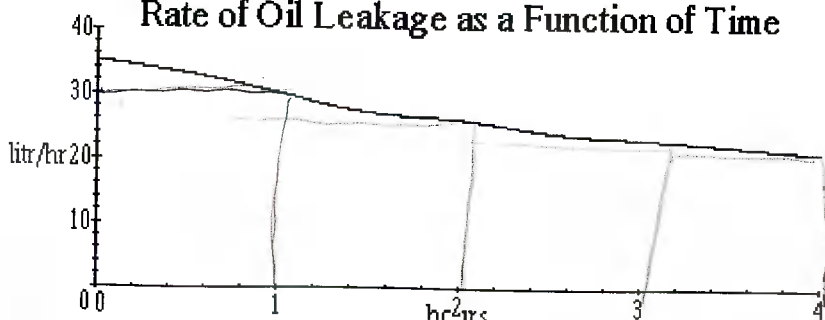
4. How accurately could you estimate the total leakage if you had access to readings of the rate of leakage at **every instant** during the four-hour interval?

depends on freq. of measurements $\rightarrow$ smaller intervals give higher accuracy
 $\rightarrow$ you could get an exact answer

Rate of Oil Leakage as a Function of Time



Rate of Oil Leakage as a Function of Time

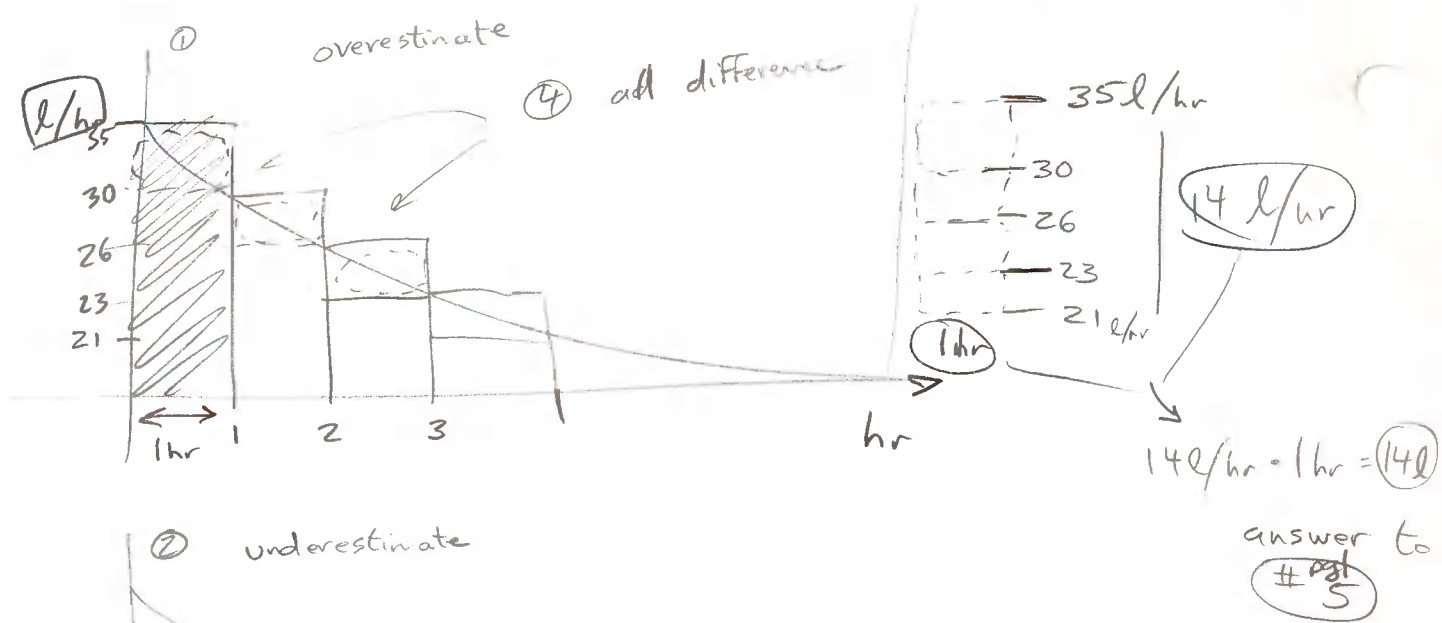


real answer is somewhere between high + low
over estimate

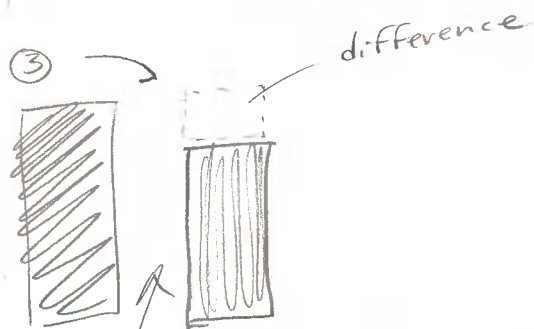
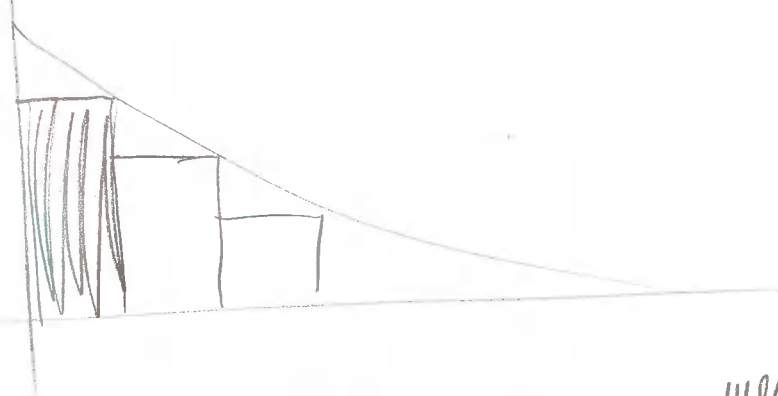
add all the difference

under estimate

$$\boxed{l/hr \cdot hr = l} \quad \text{area under the curve}$$



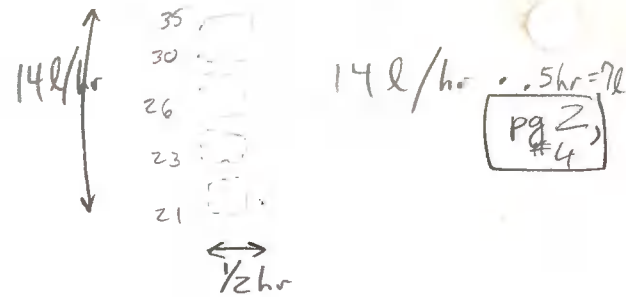
② underestimate



overestimate underestimate

right answer is
somewhere
in-between

⑤ smaller interval
will reduce
error



⑥ we can make error
as small as
we want by
reducing interval

$$14 \text{ l/hr} \cdot \frac{1}{10} \text{ hr} = 1.4 \text{ l}$$

$$14 \text{ l/hr} \cdot \frac{1}{100} \text{ hr} = .14 \text{ l}$$

$$14 \text{ l/hr} \times \text{hr} < .5 \text{ l}$$

$$x < \frac{.5}{14} \text{ hr}$$

$$x < .036 \text{ hr}$$